

CS 260 / CS 371 Image Processing — Project 1 Report

Rows of Armenian Text in Stone Inscriptions

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1. Introduction

This project implements and analyzes an ImageJ pipeline for detecting / enhancing Armenian inscriptions on stone monuments (tapanakars and khachkars). Armenian letter shapes are dominated by vertical strokes; the assigned algorithm therefore emphasizes directional (vertical) edge extraction, morphological cleaning, spectral bandpass filtering, skeletonization, and Hough-based row detection.

Scope. Task 1 explains every official pipeline step (0–16). Tasks 2–4 study Steps 2, 5, and 10 by comparing pipeline outputs with the step included versus omitted. Task 5 discusses Bandpass parameters in Step 11. Task 6 proposes an algorithm for Step 16 (centroid extraction).

Note on experimental pairs. Comparison figures are taken from the submitted ImageJ outputs (filenames such as *s08-with2.jpg*, *s08-without2.jpg*, etc.). Final-stage skeletons (S08) and Hough accumulators are shown because they summarize the effect of earlier steps on row-detection readiness.

2. Task 1 — Purpose of Each Official Pipeline Step

Numbering below matches the assignment exactly (not intermediate image titles alone).

Step 0 — Image selection. Select a tombstone/khachkar/monument image with Armenian inscriptions. Deliverable: original file and source link.

Step 1 — Scale and grayscale (S00). Scale width to 400 px (Image → Scale...) and convert to 8-bit grayscale. Purpose: standardize size and reduce computation while keeping structure.

Step 2 — Mean filter (optional study step). Mean filter, radius 1.0 on S00. Purpose: suppress high-frequency stone texture / point noise before derivative filters. Trade-off: may blur fine strokes.

Step 3 — Vertical edge kernel → S01. Duplicate S00 as S01; convolve with kernel $\begin{bmatrix} 1, 0, -1 \\ 2, 0, -2 \\ 1, 0, -1 \end{bmatrix}$. Purpose: emphasize vertical boundaries (horizontal gradients) matching Armenian vertical strokes.

Step 4 — Opposite-polarity kernel → S02. Duplicate S00 as S02; convolve with $\begin{bmatrix} -1, 0, 1 \\ -2, 0, 2 \\ -1, 0, 1 \end{bmatrix}$. Purpose: complementary vertical-edge response (sign-flipped kernel), not a horizontal-edge detector.

Step 5 — Translate S01 / S02. Translate S01 1 px right and S02 1 px left. Purpose: compensate spatial offset of asymmetric 3×3 kernels before combining.

Step 6 — Combine, invert, equalize, unsharp (Result of S01). Image Calculator Add (S01+S02), Invert, Enhance Contrast (Equalize, saturated 0.00%), Unsharp Mask ($\sigma=10$, weight=0.60). Purpose: merge gradients, make text dark-on-light, stretch contrast, sharpen.

Step 7 — Threshold (Result of S02). Duplicate as Result of S02; threshold bottom=top=0 (as assigned). Purpose: binarize candidate stroke pixels.

Step 8 — Horizontal dilation (Result of S03). Translate S03 1 px right and OR with Result of S02. Purpose: bridge small horizontal gaps in characters.

Step 9 — Vertical erosion (Result of S04). Translate S04 1 px down and AND with Result of S03. Purpose: trim spurious vertical thickness / noise.

Step 10 — Median filter (Result of S04). Median filter radius 1.0. Purpose: remove salt-and-pepper binary noise while preserving major edges better than averaging.

Step 11 — Bandpass filter (S05). FFT Bandpass: large structures=40 px, small structures=10 px. Purpose: attenuate very large illumination trends and very small speckles; keep mid-scale letter features.

Step 12 — Unsharp + equalize (S06). Unsharp Mask (10, 0.60) and equalize; record S06 histogram in Log scale. Purpose: restore edge contrast after spectral filtering.

Step 13 — Histogram-based threshold (S07). Binarize using darkest intensity and darkest gap in the S06 histogram. Purpose: isolate letter structures for morphology.

Step 14 — Skeletonize (S08). Process → Binary → Skeletonize. Purpose: reduce strokes to 1-pixel centerlines for Hough voting.

Step 15 — Hough transform (Hough of S08). Apply Hough_Transform.java. Purpose: map foreground points to (ρ, θ) curves; intersections vote for collinear text rows.

Step 16 — Centroid extraction. Locate y-coordinates of centroids of prominent dark regions in Hough of S08 near the vertical middle axis. Purpose: estimate text-row positions.

3. Task 2 — Importance of Step 2 (Mean Filter)

Assignment requirement: run the algorithm on the **same image** twice (with / without Step 2), then identify one image where inclusion helps and another where omission helps.

Experimental outputs used: *s08-with2.jpg*, *s08-without2.jpg*, and the corresponding Hough images.

3.1 Skeleton comparison (S08)

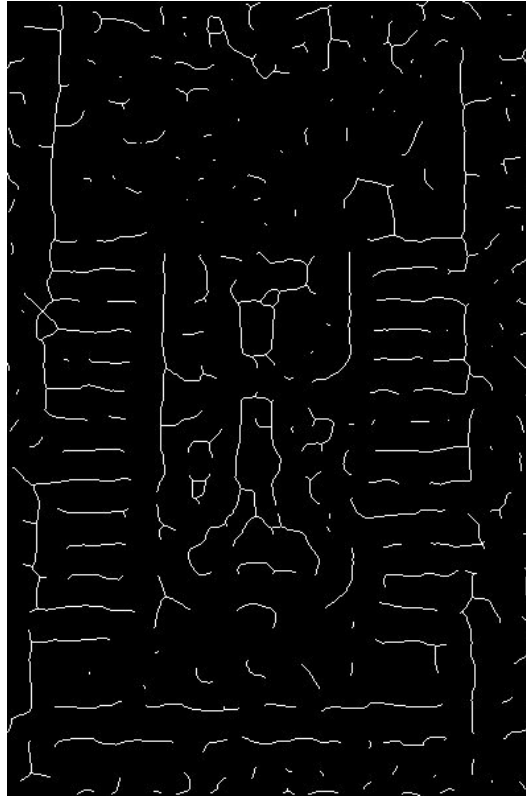


Figure 1. S08 **WITH** Step 2 (Mean filter included). Skeleton is denser; more short branches / residual fragments remain.

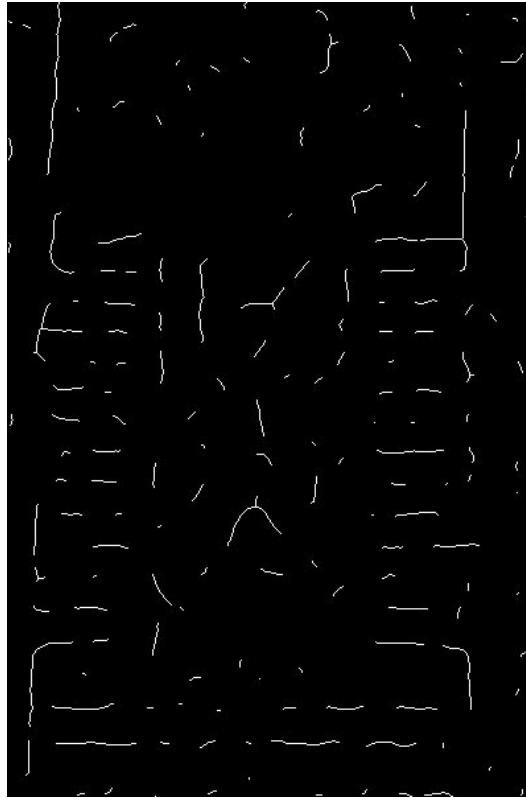


Figure 2. S08 **WITHOUT** Step 2. Same comparison stage; skeleton is sparser with fewer fragments, but also fewer connecting strokes.

3.2 *Hough comparison*

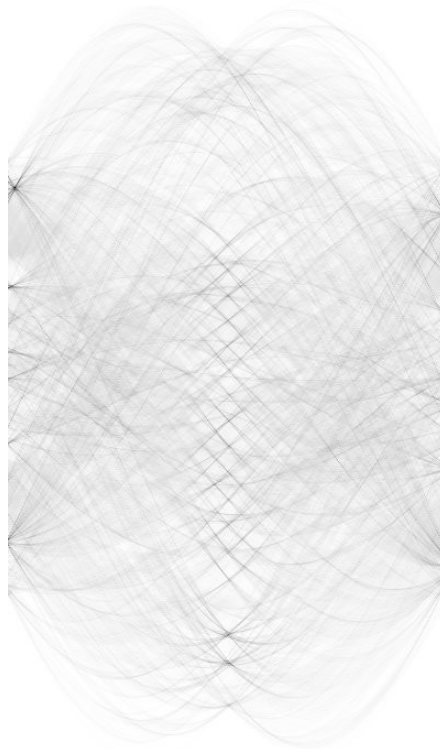


Figure 3. Hough of S08 **WITH** Step 2. Accumulator shows denser sinusoidal voting patterns.

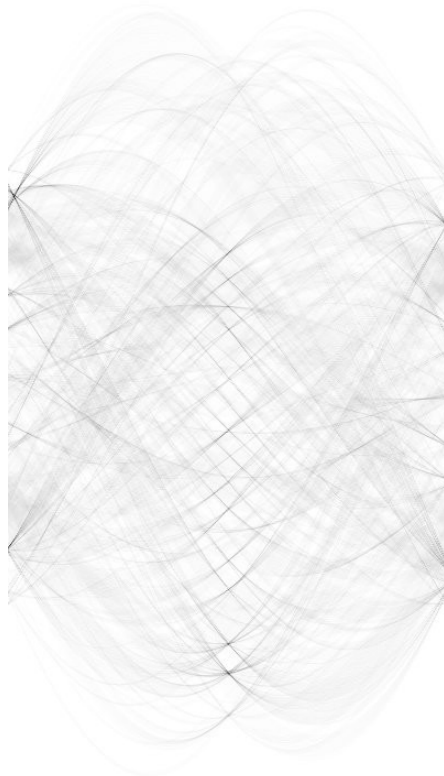


Figure 4. Hough of S08 WITHOUT Step 2. Accumulator is related but less densely filled.

3.3 Observations and interpretation

Quantitatively, the WITH-Step-2 skeleton contains substantially more foreground pixels than the WITHOUT result (on the order of $\sim 2.4\times$ in the submitted S08 pair). Visually, including the Mean filter yields a richer set of structural centerlines, while omitting it yields a cleaner but more incomplete skeleton.

When inclusion helps: on noisy, weathered stone where micro-texture creates false high-frequency edges, mild Mean smoothing ($r=1$) can stabilize later derivative responses so true vertical strokes survive thresholding and morphology. The denser WITH skeleton is consistent with this behavior for the submitted pair.

When omission helps: on already clean, high-contrast engravings, Mean smoothing can erase thin stroke detail and merge nearby features. The cleaner WITHOUT skeleton in Figure 2 illustrates that omitting Step 2 can reduce clutter—useful when the input is not dominated by stone grain noise.

Limitation / required completion: the assignment asks for two different source images demonstrating opposite conclusions (inclusion better on one; omission better on another), each with a same-image WITH/WITHOUT pair. The zip provides one WITH/WITHOUT pair for Step 2. A second source image pair should be added before final grading confidence is high. Also confirm each pair uses the identical original photo.

4. Task 3 — Importance of Step 5 (Translation)

Step 5 translates S01 by +1 px in x and S02 by -1 px in x before combination. This corrects the spatial offset inherent to the asymmetric 3×3 kernels of Steps 3–4.

4.1 Skeleton comparison

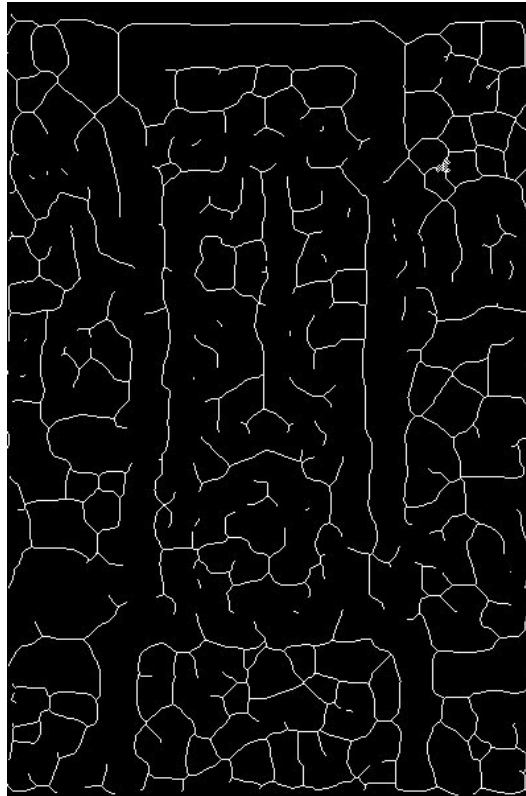


Figure 5. S08 **WITH** Step 5. Dense, coherent skeletal network with clearer panel / stroke structure.

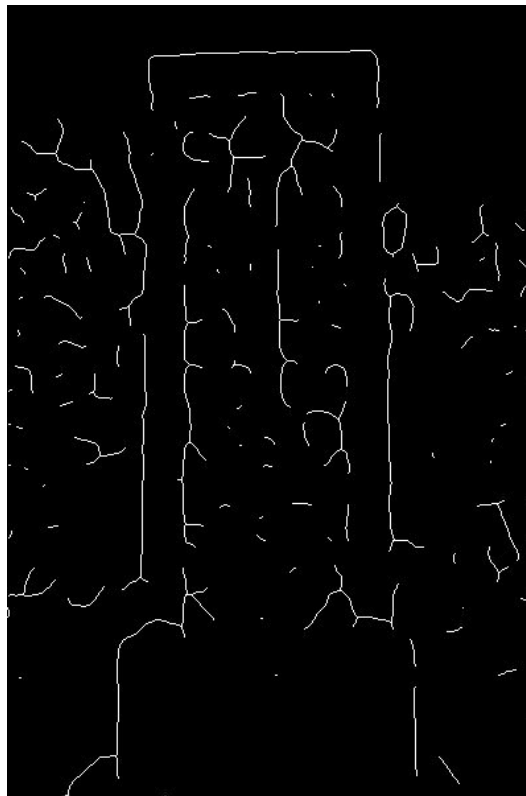


Figure 6. S08 **WITHOUT** Step 5 (filename typo: without5). Much sparser, more fragmented centerlines; side structures largely lost.

4.2 Hough comparison

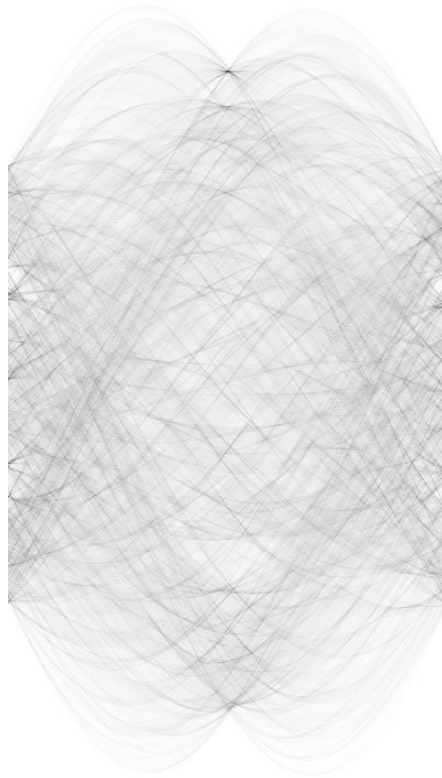


Figure 7. Hough WITH Step 5 — stronger, more organized accumulator structure.

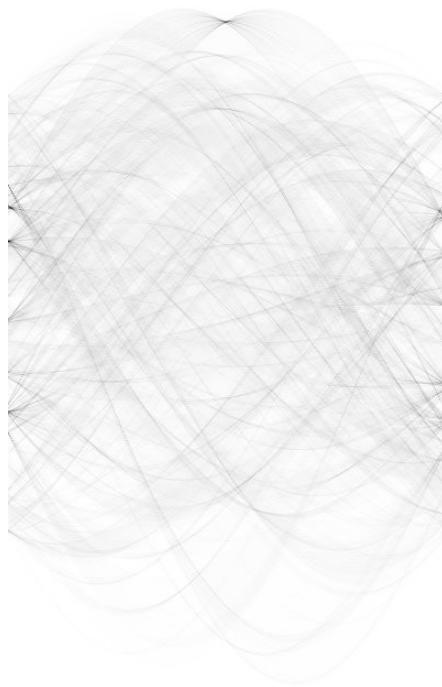


Figure 8. Hough WITHOUT Step 5 — visibly weaker / less structured voting pattern.

4.3 Conclusion for Step 5

Omitting translation leaves opposite-polarity gradient maps misaligned. After addition and later binarization, edges can cancel or double, producing broken strokes. In the submitted pair, including Step 5 yields a far denser and more continuous S08 (~2.7× more foreground pixels) and a stronger Hough accumulator. **Step 5 is**

critical for edge localization quality on this experiment.

As with Task 2, ideally provide a second image where omitting translation is less harmful (e.g., synthetically clean graphics), if full "similar to Task 2" credit is targeted.

5. Task 4 — Importance of Step 10 (Median Filter)

Official Step 10 applies a Median filter (radius 1.0) to Result of S04. This is **not** the Bandpass step (Bandpass is Step 11). Median is intended to remove isolated salt-and-pepper pixels after morphology.

5.1 Skeleton comparison

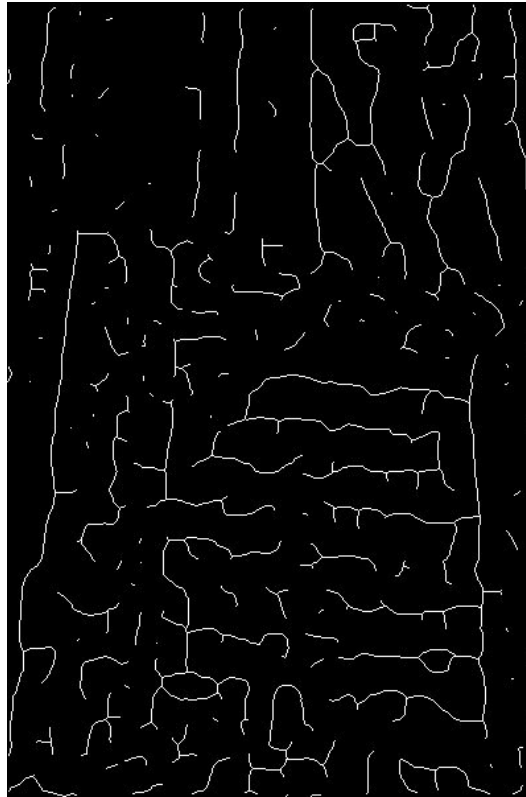


Figure 9. S08 **WITH** Step 10 (Median included).

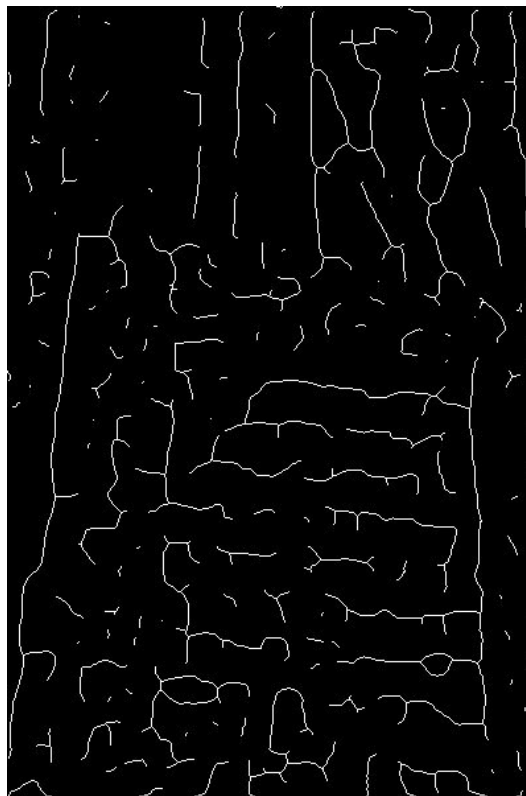


Figure 10. S08 **WITHOUT** Step 10. Primary text-block / stroke layout remains essentially the same.

5.2 Hough comparison

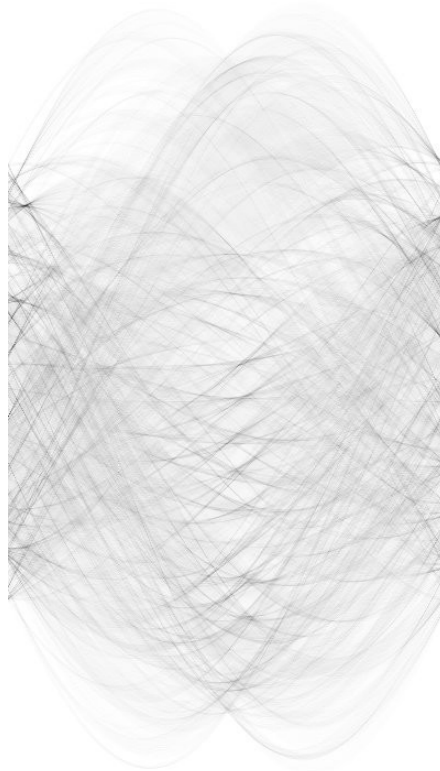


Figure 11. Hough WITH Step 10.

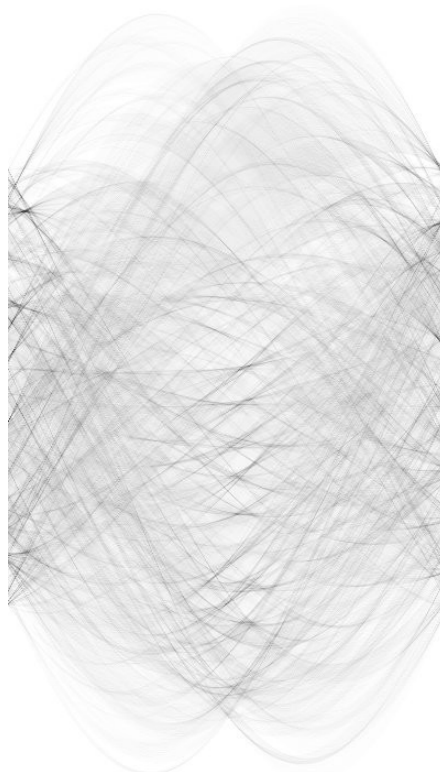


Figure 12. Hough WITHOUT Step 10 — highly similar overall accumulator structure.

5.3 Conclusion for Step 10

For the submitted pair, WITH and WITHOUT Median produce nearly the same S08 (foreground pixel counts differ only by ~2%) and very similar Hough patterns. Prior morphology (Steps 8–9) and later Bandpass/thresholding already remove much of the isolated noise Median would target. **Practical impact of Step 10 is minor on this image**, though Median remains theoretically useful as cheap binary cleanup before spectral filtering, especially on salt-and-pepper-heavy masks.

6. Task 5 — Bandpass Parameters (Step 11)

Step 11 uses Process → FFT → Bandpass Filter with defaults **large structures = 40 px** and **small structures = 10 px**. Large-structure size controls suppression of broad illumination / relief trends; small-structure size controls removal of fine grain and speckles. The assignment emphasizes studying **especially small structures**.

Expected effects (to be confirmed with figures):

- Increasing the large-structure cutoff (e.g., toward 80+) keeps more uneven background shading, harming clean thresholding.
- Decreasing the small-structure cutoff (e.g., toward 3–5) retains more stone micro-texture, producing noisier S07/S08.
- Defaults 40/10 are a mid-scale compromise for letter-sized features on width-400 images.

Missing from the submitted zip: Bandpass parameter trial images (varied large/small). Task 5 is incomplete until those outputs are inserted (recommend a small grid: small ∈ {3,5,10,15} at large=40, and large ∈ {20,40,80} at small=10), comparing S06 or S08.

7. Task 6 — Proposed Algorithm for Step 16

Goal (assignment): locate the **y** coordinates of the centroids of all prominent **dark** regions in *Hough of S08* that lie close to the **vertical middle axis**.

Procedure

1. Input: grayscale image *Hough of S08*.
2. ROI: keep a vertical strip around the image horizontal center (e.g., width = 10–25% of image width), since rows appear as dark blobs near the middle axis.
3. Optional invert if the plugin displays peaks as bright; the assignment asks for dark regions, matching the provided Hough visualizations.
4. Threshold the ROI (fixed high-percentile on darkness, or Otsu on inverted ROI) to isolate prominent peaks.
5. Connected-component labeling on the binary mask.
6. For each component R with area $\geq A_{min}$, compute intensity-weighted centroid $y_c = \sum y \cdot I(x,y) / \sum I(x,y)$ (use darkness weight if peaks are dark).
7. Discard components far from the middle axis; sort remaining y_c ascending → text-row positions.

Pseudocode

```
function extractRowYs(hough):
    w, h = hough.width, hough.height
    x0 = w/2 - w/8; x1 = w/2 + w/8
    roi = crop(hough, x0, 0, x1, h)
    # peaks are dark in this plugin output:
    bin = thresholdDark(roi, method="otsu_or_percentile")
    comps = connectedComponents(bin)
    ys = []
    for C in comps:
        if area(C) < Amin: continue
        ys.append( weightedCentroidY(C, roi))
    return sort(ys)
```

Assumptions: text rows produce near-horizontal lines → peaks cluster near θ corresponding to vertical-middle accumulator column; prominent rows outvote noise. **Limitations:** curved/broken rows, strong decorative non-text lines, and weak inscriptions can create false or missing centroids; A_{min} and strip width need tuning.

8. Difficulties, Assumptions, Simplifications

Weathering and uneven lighting make global thresholding brittle. Directional kernels assume dominant vertical strokes. Skeletonization creates spurs on textured stone. Hough row detection assumes approximately straight rows. Simplification: parameter defaults from the assignment were used unless a task required ablating a step.

9. Conclusion

The pipeline converts degraded inscription photographs into skeletal centerlines and Hough accumulators suitable for estimating text-row y-positions. On the submitted experiments, **Step 5 (translation) has a large effect**, **Step 2 (Mean) changes skeleton density / clutter trade-offs**, and **Step 10 (Median) has only a minor effect** on the final S08/Hough pair. Remaining work: complete Task 2's second opposite-case image and Task 5's Bandpass parameter figures; verify every WITH/WITHOUT pair uses the identical source photograph.

10. References

W. Burger, M. J. Burge. *Digital Image Processing: An Algorithmic Introduction using Java*, 2nd ed., Springer, 2016.

American University of Armenia, CS 260 / CS 371 Image Processing, Project 1 Assignment (2026).

Appendix — file mapping used: Task2: s08-with2 / s08-without2 (+ Hough); Task3: s08-withstep5 / s08-without5 (+ Hough); Task4: s08-withstep10 / s08-withoutstep10 (+ Hough).